

AAYUSH KUMAR JVS

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SUMMARY

- AI/ML and Full-Stack Engineer with 5+ years of experience including leading GIS systems at Apple Maps and web/cloud development. Experienced in delivering scalable, production-grade solutions using Vue, AWS, Python, and CI/CD pipelines. Published researcher with specialization in explainable AI.

EXPERIENCE

- Quebec AI Institute**
Software Developer (Quebec, CA) Jan 2023 – Dec 2024
 - Full-Stack Dev** Built a website using **Vue.JS**, **NUXT**, **AWS**, and **GraphQL**.
 - Cloud Optimization** Managed **AWS Lambda**, **S3**, and **DynamoDB**, improving cost-efficiency by 20%.
 - AI Research** Led research on Quantized LLMs for mental health. Published in **ProQuest LLC**.
- Google Summer of Code**
Software Engineer May 2023 – Sept 2023
 - Google-Sponsored Project** Worked on a Google-sponsored project in collaboration with Red Hen Labs, contributing to the development of **Red Hen Annotator Tool v2.0**.
 - Tool Enhancement** Enhanced and expanded the capabilities of the initial version developed by previous contributors, improving its functionality and usability.
 - Technology Stack** Built an end-to-end audio annotation system leveraging **NVIDIA's Guardrails** framework.
- Naresh IT**
Software Engineer May 2023 – May 2024
 - Software Development** Worked as a Software Engineer, contributing to the development, maintenance, and enhancement of software applications and internal systems.
 - Application Development** Designed and implemented application features using **Python**, **JavaScript**, and **web technologies**, with a focus on functionality, reliability, and maintainability.
 - Problem Solving** Debugged software issues, optimized application workflows, and collaborated with team members to deliver solutions within project requirements and timelines.
- RMSI Pvt Ltd**
Software Engineer (GIS) July 2020 – August 2021
 - GIS Platform Development** Built a full-stack GIS tool integrated with Confluence and Jira, improving spatial data ops by 35%.
 - CI/CD Engineering** Designed CI/CD pipelines for Apple Maps' tools, accelerating deployments by 40%.
 - Team Leadership** Mentored 20+ engineers on production workflows, reducing MTTR by 30%.
 - API + SDK Integration** Enhanced GIS tools using internal REST APIs, ArcGIS, SDKs, improving ingestion times by 35%.

PROJECTS

- Encrypted Peer-to-Peer File Sharing System**
Research Project Jan 2023 – Mar 2023
 - Security Protocols** Built a decentralized encrypted P2P file sharing system using Python sockets and cryptographic primitives.
- Analysis of Secured Federated Learning**
Research Project Apr 2023 – Jun 2023
 - Federated Privacy** Explored tradeoffs between performance, privacy, and latency in secure aggregation protocols across devices.
- Master's Thesis – Quantized LLMs on Mental Health Datasets**
Research Project Feb 2024 – Nov 2024
 - Model Benchmarking** Benchmarked quantized LLMs (QLoRA, BitsAndBytes, GGUF, GGML) using IMHI datasets to evaluate tradeoffs between compression and accuracy.

SKILLS SUMMARY

- Programming Languages** Java, Python, JavaScript, TypeScript, Go, HTML5, CSS3, Wordpress
- Frameworks and Libraries** PyTorch, VueJS, React Native, Angular, NodeJS, NextJS, Django, TensorFlow, Keras
- Cloud and DevOps Tools** AWS (EC2, S3, Lambda, ECS, RDS), Microsoft Azure, Docker, Kubernetes, CI/CD, Git, Apache Kafka
- Databases and Query Languages** PostgreSQL, MySQL, MongoDB, Redis, Oracle, HiveQL, GraphQL
- Data Processing and Analytics Tools** Pandas, NumPy, PySpark, Hadoop, FusionX, ArcGIS
- Data Visualization Tools** Power BI, Tableau, D3.js, Plotly, GIS Mapping
- Development Methodologies** Agile, Scrum, Test Driven Development, DevOps, Root Cause Analysis

EDUCATION

- Master of Science - Computer Science**
The University Of Montreal; Quebec AI Institute GPA: 3.5 Aug 2021 – Dec 2023
- Bachelor of Technology - Information Technology**
SRM Institute Of Science And Technology GPA: 3.0 Jul 2016 – Jun 2020